

Fluorescence

Y25 (2025) — English translation

Equipment

Electrical: Photosensitive CCD ruler connected to a computer, magnetic screen, USB flash drive, USB lamp.

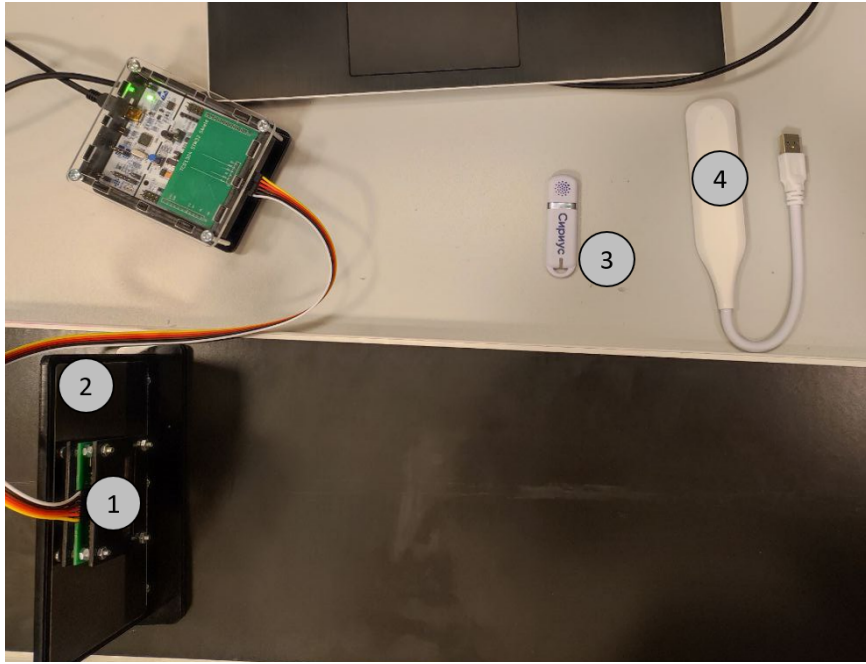


Figure 1: Electrical equipment.

Optical: Power supply (15 V), cable adapter, continuous spectrum LED (LED-CON), three-color LED (LED-RGB), three black lenses L_1 (+6 D), one white lens L_2 (+16 D), lens L_3 on a rod, diffraction grating ($d_0 = 2.0 \mu\text{m}$), optical table, stand for illuminator and cuvettes, spacer, stand for lens L_3 and laser, ruler, slit.

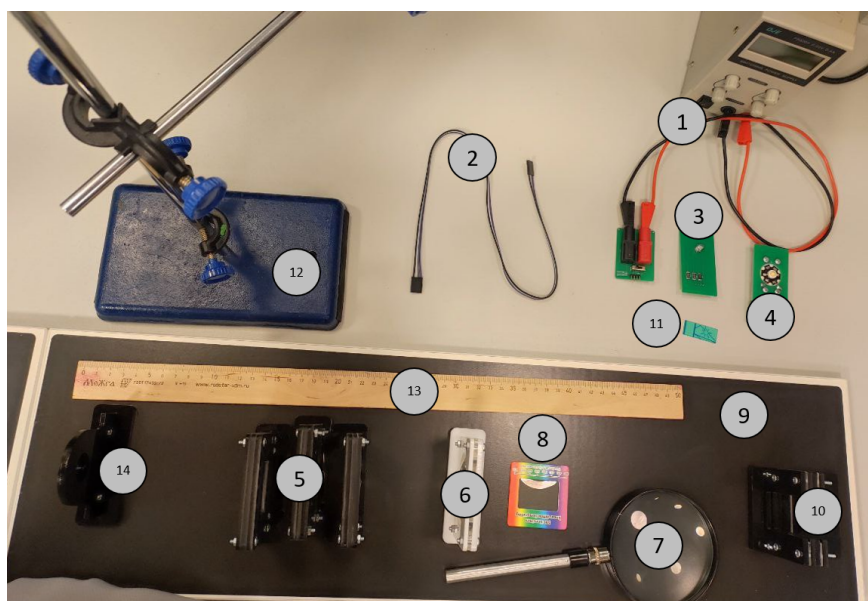


Figure 2: Optical equipment.

Chemical: 10 cuvettes with lids, analysis jars with water, 0.5 ml syringe, two 3 ml syringes, red/green/blue lasers, batteries, laser button holder, Rhodamine B solution ($c_0 = 200 \mu\text{M}$), tweezers.

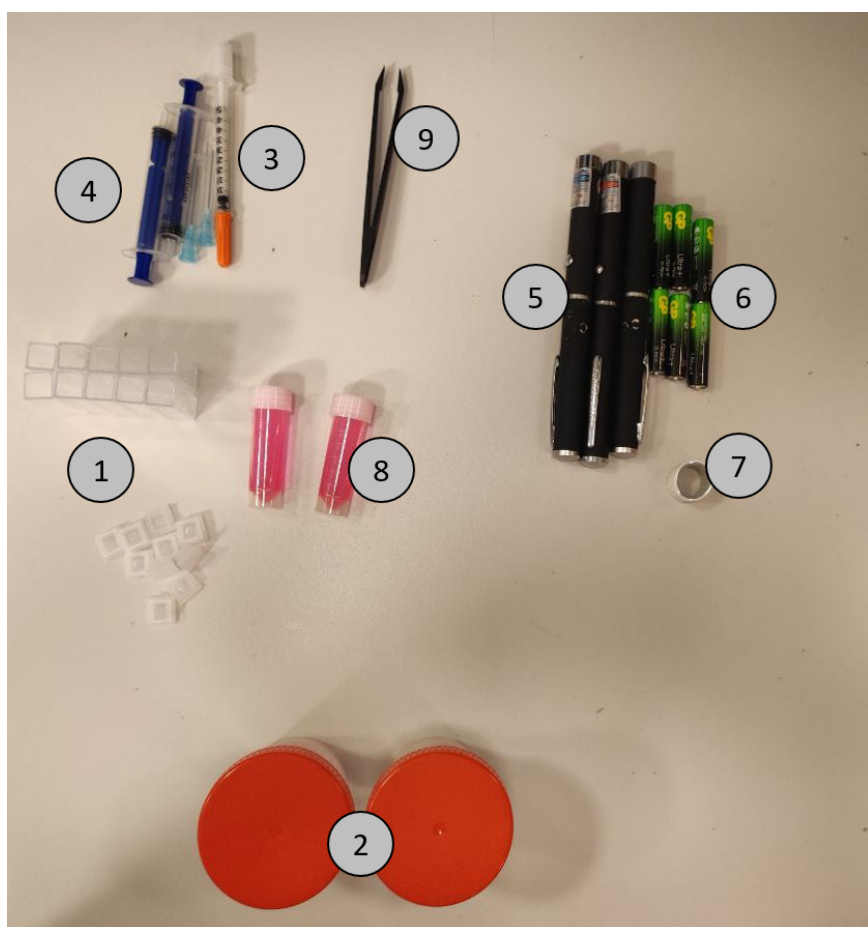


Figure 3: Chemical equipment.

Introduction

The Jablonski diagram is used to represent energy states. For Rhodamine B (RhB), the total energy E is the sum of electronic energy E_e and atomic vibration energy E_a ($E_a \ll E_e$). S_0 is the ground state (energies set to zero). S_1 is the first excited electronic state. Vibrational levels are denoted by ν .

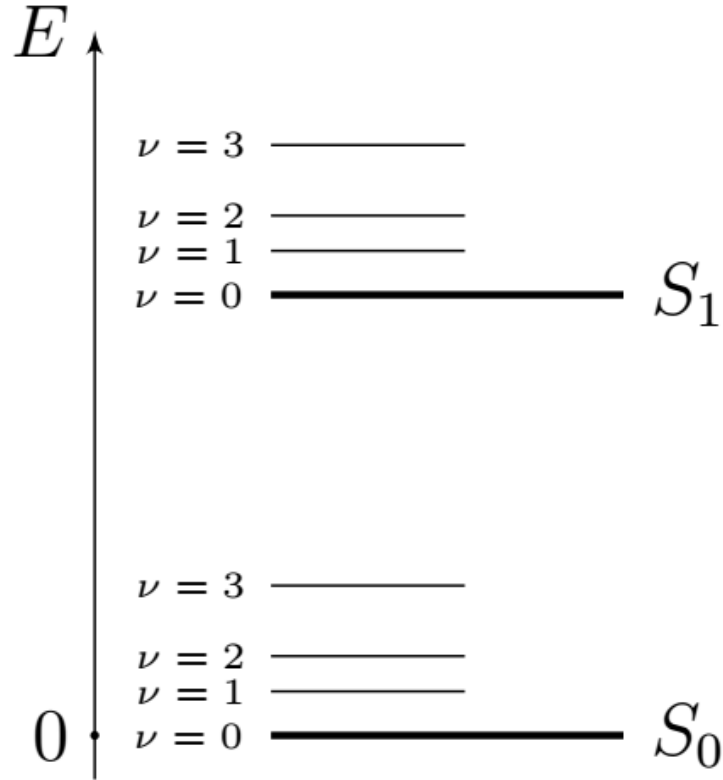


Figure 4: Jablonski diagram levels.

The photon energy is $\hbar\omega$. If a molecule is in $S_0, \nu=0$, it can absorb a photon if $\hbar\omega = E_1 - E_2$. Absorption and emission lines are broadened; the width is characterized by the FWHM (Full Width at Half Maximum).

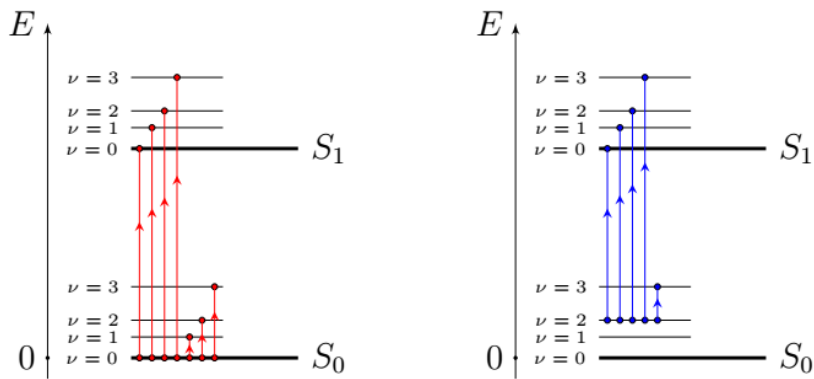


Figure 5: Absorption processes from $S_0, \nu=0$ and $S_0, \nu=2$.

Constants: $\hbar = 1.05 \cdot 10^{-34}$ J·s, $c = 3.00 \cdot 10^8$ m/s, $e = 1.60 \cdot 10^{-19}$ C, $N_A = 6.02 \cdot 10^{23}$ mol⁻¹.

Part A. Absorption spectrum (0.6 points)

A1	Draw a qualitative Jablonski diagram for perylene. Indicate the values of the energies of the states. Take the ground state energy equal to zero.	0.30
A2	Based on the infrared spectrum (Fig. 9), indicate the transition corresponding to the absorption maximum.	0.30

Part B. Absorption spectroscopy (8.5 points)

B1	Determine the focal length of the lens L_3 .	0.20
B2	Prepare a solution of RhB with concentration $c = 50$ μ M. Qualitatively indicate which color of light is absorbed most strongly in it.	0.20

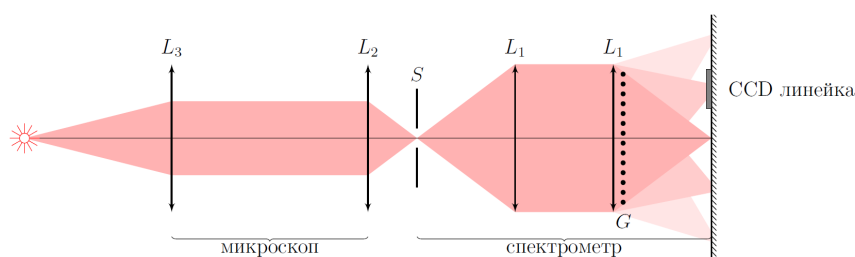


Figure 6: Optical scheme for spectrum measurement.

B3	Adjust the positions of all lenses and the CCD ruler so the first-order diffraction of visible light falls on the CCD. Measure the spectra $I(\lambda)$ for the RGB-LED colors. Save as R.csv, G.csv, B.csv.	0.30
B4	For each RGB color, indicate the pixel number $N_{\max}$ corresponding to the peak emission.	0.30
B5	Establish the calibration $\lambda = \text{slope} \cdot N + \text{shift}$ and record your values for slope and shift.	0.20
B6	Find the pixel pitch value Δx of the CCD ruler.	0.40
B7	Obtain the spectrum $I_0(\lambda)$ for LED-CON through a water cuvette. Save as LED-CON.csv.	0.10

B8	Obtain the spectrum $I(\lambda)$ of LED-CON light after passing through 50 μM RhB. Save as T-50.csv.	0.40
B9	Plot $I(\lambda)/I_0(\lambda)$ and determine the wavelength $\lambda_{\text{RhB,max}}$ where absorption is maximum.	0.30
B10	Measure $I(\lambda)$ for 6 other concentrations in the range $[1, 100]$ μM . Save as T-c.csv.	2.00
B11	Determine the ratio I/I_0 for a solution layer of thickness L and concentration c in terms of the molecular cross-section σ .	0.30
B12	Plot $\ln(I_0/I)$ for $\lambda = \lambda_{\text{RhB,max}}$ versus c .	0.90
B13	For a mixture of monomers and dimers, determine I/I_0 in terms of $\sigma_m, c_m, \sigma_d, c_d$ and L .	0.30
B14	Identify concentrations where $c_m \approx c$ from your plot.	0.20
B15	Derive the formula for wavelength-to-energy conversion: $E/\text{eV} = A \cdot (\lambda/\text{nm})^\gamma$.	0.10
B16	Plot $\sigma(E)$ for RhB monomers.	1.50
B17	Draw the qualitative Jablonski diagram for RhB using energies in eV.	0.30
B18	Indicate the vibrational transition for the IR absorption maximum in Fig. 13.	0.30

Part C. Fluorescence microscope (10.9 points)

Fluorescence involves absorption ($S_0 \rightarrow S_{1,\nu}$), non-radiative relaxation to $S_{1,0}$, and spontaneous emission ($S_{1,0} \rightarrow S_{0,\nu}$). The Stokes shift describes the shift between absorption and emission maxima.

C1	Describe the observed phenomena when shining the red, green, and blue lasers on the RhB solution. Explain using your Jablonski diagram.	0.30
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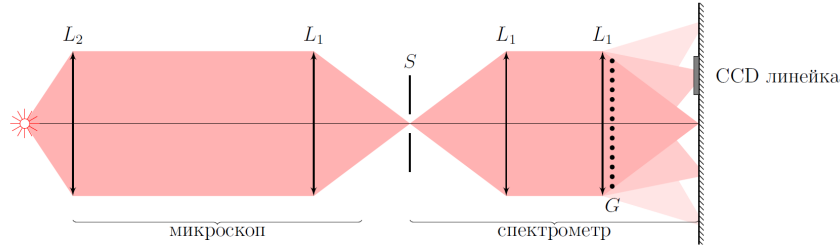


Figure 7: Modified optical design for fluorescence.

C2	Recalibrate the positions using the RGB-LED and save the spectra (R.csv, G.csv, B.csv) in folder C2.	0.30
C3	Express the average time τ between photon hits in terms of σ , I_0 , and ω_0 .	0.20
C4	Estimate the focus intensity I_0 and the time τ for the blue laser ($P_0 = 5$ mW, $d_0 = 2$ mm).	0.90
C5	Express the fluorescent power P per unit volume in terms of σ , I_0 , τ , τ_0 and c .	0.40
C6	Obtain the RhB fluorescence spectrum (100 μ M) under blue laser excitation. Save as 405.csv and find $\lambda_{405, \max}$.	0.50
C7	Study $I_{405}(\lambda_{405, \max})$ as a function of the volume V of solution added. Save as 405-V.csv.	2.30
C8	Plot the linearized graph and determine the absorption cross-section σ_{405} .	1.50
C9	Measure fluorescence spectra for four other concentrations c . Save as 405-c.csv.	2.00
C10	Plot $I_{405}(\lambda_{405, \max})$ versus c .	1.20
C11	Obtain and save (532.csv) the fluorescence spectrum under green laser excitation.	0.20
C12	On one graph, plot normalized absorption $A(E)$ and fluorescence $F(E)$.	0.80
C13	Draw the final qualitative Jablonski diagram for RhB.	0.50

Source: <https://pho.rs/p/4541>